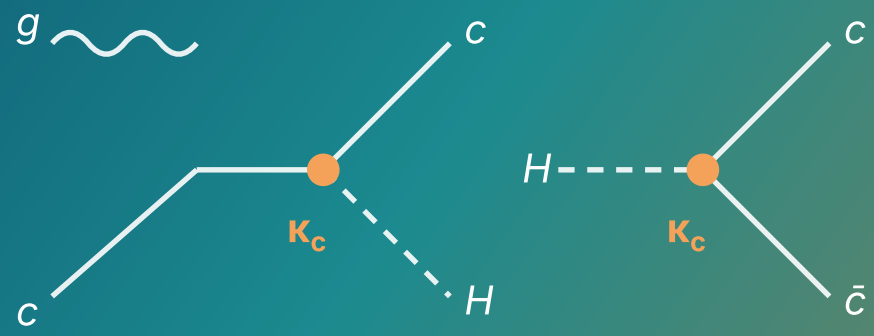
# Constraints on Higgs–charm couplings at CMS

*Higgs 2026 · Louvain-la-Neuve · 28 October 2026*

**Chirayu Gupta**
*Vrije Universiteit Brussel*

on behalf of the CMS Collaboration



CMS-DP-2025-081

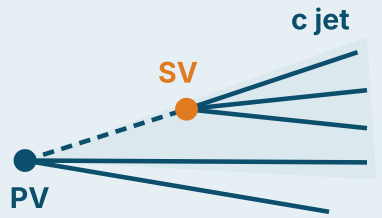
# Charm tagging

## Finding c jets between b and light jets

### Charm signature

c-hadron mass and lifetime lie **between light and b**

- secondary vertex (SV), like b jets
- but closer to the primary vertex: **easily mistaken for light jets**



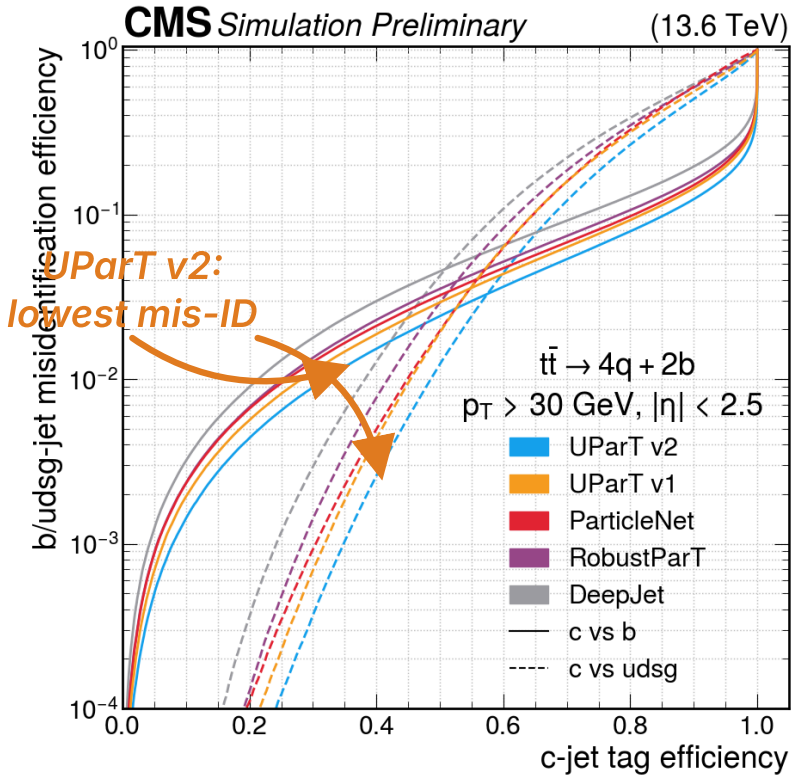

### Algorithm evolution at CMS

**Likelihood / BDT**
JP, CSV:
tracks + secondary vertices

**Deep NN**
**DeepJet**:
all jet constituents

**Particle cloud**
**ParticleNet**, UParT:
graph, transformer

*Run 1 → Run 3: richer networks, inputs closer to raw particles*




t t̄ simulation, 13.6 TeV · solid: c vs b, dashed: c vs light

**In this talk** DeepJet · VH resolved, cH | ParticleNet · VH merged, t t̄H | DeepDoubleX · boosted H

# VH(H → $c\bar{c}$)

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## Cleanest signature: the W or Z decays to leptons

**0ℓ · Z → νν** **1ℓ · W → ℓν** **2ℓ · Z → ℓℓ**

V+jets and t $\bar{t}$ normalised in control regions

**$p_T^H$ = 300 GeV**

**Resolved · two jets**

>95% of the signal
two DeepJet c-tagged jets
fit to a BDT score

*below*

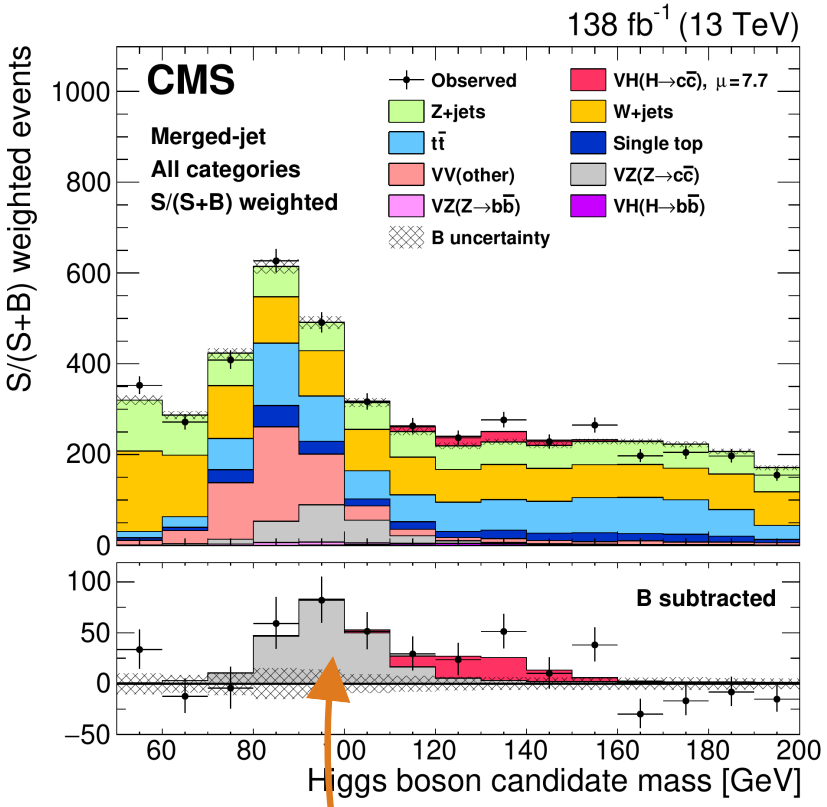
**Merged · one large jet**

<5% of the signal, **much less background**
ParticleNet $c\bar{c}$ tagging
fit to the regressed mass

*above*

95% CL · observed (expected)

**$\mu$ < 14** (7.6) **1.1 < $|\kappa_c|$ < 5.5** (< 3.4)




Merged-jet candidate mass, background-subtracted below

*VZ(Z → $c\bar{c}$) seen*

***➜ Most sensitive single channel of Run 2 · statistics-limited***